Target Indices After Sorting Array

Problem

You are given a 0-indexed integer array **nums** and a target element **target**. A target index is an index **i** such that **nums[i] == target**. Return a list of the target indices of **nums** after sorting **nums** in non-decreasing order. If there are no target indices, return an empty list. The returned list must be sorted in increasing order.

Examples

Example 1:
Input: nums = [1,2,5,2,3], target = 2
Output: [1,2]
Explanation: After sorting, nums is [1,2,2,3,5]. The indices where nums[i] == 2 are 1 and 2.

Example 2:
Input: nums = [1,2,5,2,3], target = 3
Output: [3]

Example 3:
Input: nums = [1,2,5,2,3], target = 5
Output: [4]

Approach

Sort the array in non-decreasing order. After sorting, all occurrences of the target appear consecutively. Scan the sorted array once and add every index whose value equals the target. Because the scan moves left to right, the resulting indices are automatically in increasing order.

Java Solution

```
import java.util.ArrayList;
import java.util.Arrays;
import java.util.List;

class Solution {
    public List<Integer> targetIndices(int[] nums, int target) {
        Arrays.sort(nums);

        List<Integer> result = new ArrayList<>();

        for (int i = 0; i < nums.length; i++) {
            if (nums[i] == target) {
                result.add(i);
            }
        }

        return result;
    }
}
```

Complexity

Time complexity: O(n log n), due to sorting.
Space complexity: O(1) auxiliary space, excluding the output list.